Quality Without a Single Right Answer: Separating Shared Quality, Personal Taste and Noise in Judgments of Language-Model Responses

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Abstract. Most requests to language models have no single correct output, yet choosing among models needs a measure of answer quality. We assemble standard results into a measurement model for the quality of one response that separates correctness, shared quality, personal taste, leniency and noise. The mathematics is not new; it shows which designs measure which component, defines a subjectivity index S, and yields an accuracy-first rule that admits a cheaper model only if non-inferior. In a controlled study of 90 tasks from two documentation sources, with two fact checkers and two simulated rater families, a cheaper model matched a stronger one on correctness yet won only 17 to 42% of comparisons (ties as half). Crossed ratings with repeats estimated every component, leniency and noise included. S was lowest for documentation questions under both families, but its size depended on the family (0.16 to 0.38 or 0.40 to 0.75), and pairwise judgments pinned it down only weakly; these findings concern estimators, not people. In MT-Bench's 3,355 expert votes, experts prefer the correct answer in 89 to 99% of votes where only one is correct. Opus 5.5 agrees with an expert more often than another expert does, but it reproduces the benchmark's questions and the other experts' majority agrees more often still. On PRISM, a person's own taste improves prediction by 0.7% of variance, an upper bound because participants chose their prompts. Leniency, noise and within-rater cycles remain untested on people; we size the panel that would test them.

Keywords: evaluation of language models · subjective quality · preference measurement · generalizability theory · Hodge decomposition · personal taste · model selection

1 Introduction

Benchmarks measure what can be checked: whether code passes its tests, whether a number is right, whether a statement matches a source. Most requests that people actually send are not of that kind. "Reply to this angry customer", "write the launch post" and "what should we do about our churn?" admit many acceptable answers, and which one is best depends partly on the answer and partly on

who reads it. Such requests are routinely called *subjective*, and the label usually ends the analysis. For anyone choosing among models, or replacing an expensive model with a cheaper one, this is the regime that matters most and is measured least.

This paper describes such requests with an explicit measurement model rather than the label. Its starting point is that a judgment of an answer is a function of who judges, what is judged and when. Classical results on functions of independent variables [21,18], the population form of the variance-components models of generalizability theory [13,9], then split any judgment exactly and orthogonally into what all raters agree on about the answer (*shared quality*), how lenient a rater is, what a particular rater likes about a particular answer (*personal taste*), and what changes when the same judgment is repeated (*noise*). Projecting the shared and personal parts onto checkable properties and onto style features yields correctness, shared style and taste; in pairwise form the result is a random-coefficients (mixed) logit model [35]. For pairwise judgments, the Hodge decomposition [24] isolates the one structure that no utility-based model can represent: preference cycles within a rater. Figure 1 summarises the model.

None of this mathematics is new, and we do not claim it as a contribution. Section 4 restates standard results in the vocabulary of response quality, with their sources. What the paper adds is the assembly of these results into one definition of the quality of a single response, the design requirements that follow from it, and an empirical study that reports, component by component, what the available human data can and cannot establish.

Contributions.

1. A measurement framework for the quality of a response to a request without a single right answer, in which correctness, shared quality, personal taste, leniency, noise and within-rater cycles are orthogonal components of one identity, assembled from standard results (Sections 3 and 4).
2. Design consequences: which data designs identify which component (Proposition 3), a subjectivity index that is invariant to leniency and noise, an accuracy-first decision rule for model selection with the sample sizes it implies (Section 5), and the size of a human panel that would test every component, computed from variance components measured in the controlled study (Section 6.6).
3. A controlled study on 90 tasks from two documentation sources, one unrelated to the author, with two independent fact checkers, raters portrayed by two model families, and both a pairwise and a crossed rating design, so that every component of the model, leniency and noise included, is estimated at least once (Section 6.1). It also shows two points about measurement: when most choices are unanimous, pairwise judgments pin down the subjectivity index poorly, so crossed ratings with repeats are needed to estimate it; and its size depends on which model portrays the raters, although documentation questions have the lowest index under both.

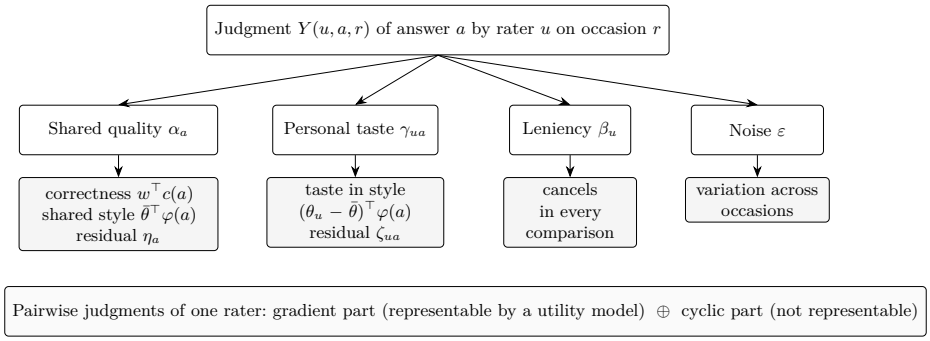


Fig. 1. The measurement model. Every judgment splits exactly and orthogonally into four components (Theorem 1). The shared and personal components split further by projection onto checkable properties c and style features φ (Theorem 2). For pairwise judgments, the Hodge decomposition separates what any utility model can represent from within-rater cycles (Theorem 3).

4. Evidence from human judgments: MT-Bench's expert votes [53] and PRISM [28] (Sections 6.2 and 6.3). Table 6 records which components each source can test. On human data, correctness and shared quality are supported, personal taste is detected but small and possibly confounded with the choice of prompts, and leniency, noise and within-rater cycles are not testable with these designs.
5. Checks on the instruments: a second correctness grader, data-only baselines and contamination probes for the comparison of a model judge with experts, and a calibration check of the cycle test.

Variance decompositions have been used to assess the reliability of benchmarks scored by LLM judges, with systems, items and judges as facets [2,55,36]; heterogeneous preferences have been modelled for training [49,44,50]; and the intransitivity of LLM judges has been measured [52]. Our object differs: the quality of a single response for a population of raters, with that population's disagreement treated as part of the quantity rather than as error.

2 Related work

Pairwise preference evaluation. Chatbot Arena fits Bradley-Terry scores [8] to crowdsourced blind votes [12]. Zheng et al. [53] showed that strong LLM judges agree with human preferences at about the rate at which humans agree with each other, and documented position, verbosity and self-enhancement biases. Length-controlled AlpacaEval removes the effect of length from judge preferences by regression [17], and Arena's style control adds style features to the Bradley-Terry fit [30]. Arena rankings can change when a tiny fraction of votes is removed [23]. GDPval measures professional deliverables by blinded expert comparison with

human work [41]. Our shared-quality term with style features generalises style control, and our personal term is what pooled Bradley-Terry fits average away.

Reliability and variance components. Generalizability theory decomposes score variance into facets and their interactions [13,9]. Recent work applies it to LLM evaluation: Anand [2] shows that a system-by-judge interaction bounds the reliability of single-judge pointwise scoring, Zhu et al. [55] rotate judges to average out judge bias, Messing [36] shows that standard confidence intervals ignore variance due to design choices, and Zhu [54] shows how classical reliability statistics can mislead about LLM judges. Sadab and Sakhawat [46] make the point with human ratings of eight systems: a conventional analysis declares 19 of 28 pairwise differences significant, while treating annotators as a random factor finds no significant difference between the systems, and annotator identity explains more variance than system identity. These works study how reliably a *benchmark ranks systems*. We study what the *quality of a response* consists of, with the rater population as the object of interest rather than a nuisance facet.

Psychometrics, choice models and annotator models. Our model is close to several established ones. Many-facet Rasch measurement models a rating through an item parameter, a person parameter and a rater severity [32]; our shared quality and leniency play the roles of the item parameter and the rater severity. In choice modelling, utilities with coefficients that vary across decision makers form the mixed (random-coefficients) logit, which can approximate any random-utility model [35], a stronger expressiveness result than our Corollary 1; our personal style weights θ_u are such random coefficients. Our subjectivity index is a ratio of variance components of the kind generalizability theory uses for dependability coefficients [9]. Annotator models estimate each annotator's error rates around a latent true label [15,42]. They assume that disagreement is error around one truth, which in our terms means $S = 0$; our model lets the target itself vary by rater. In people, intransitive preferences have been demonstrated under specially constructed conditions [51].

Disagreement and pluralism. Perspectivist NLP treats annotator disagreement as signal rather than noise [43,10]. Vicarious annotation, in which raters also say how they think others would label an item, separates a rater's own view from their picture of the group [39], and the CoMeDi shared task made the amount of disagreement itself a prediction target, which proved much harder than predicting the median judgment [48]. Distributional preference learning shows that standard preference learning implicitly aggregates hidden context by a Borda count [49], and variational preference learning personalises reward models with a latent variable per user [44]. Sorensen et al. [50] set out a roadmap for pluralistic alignment, and PRISM links individual ratings to participant profiles [28]. Our personal term is the measurement counterpart of these modelling efforts, and our subjectivity index quantifies how much of a task's quality depends on the person.

Checkable decomposition. FActScore splits long answers into atomic facts checked against a source [37]. Rubric-based evaluation turns open-ended quality into weighted checklists [5,20], and rubric-following judges score against them [27]. These are instances of our correctness projection $w^\top c$.

Intransitivity. Combinatorial Hodge theory decomposes pairwise comparison flows into gradient, curl and harmonic parts [24]. LLM judges exhibit non-transitive preferences [52], and counts of circular triads [26] have been used to audit LLM rankings; Pokharel et al. [45] read their triad-based consistency coefficient as a summary of the curl component. We use the decomposition for a different purpose: to delimit what any utility model can represent, and to separate individual cycles from cycles produced by aggregating different tastes.

Judges as instruments, and routing. LLM judges favour their own generations, and the bias grows with their ability to recognise them [40]. JudgeProfile separates what a judge perceives in two responses from how much weight it gives each attribute, across 21 judges and 87 attributes [11]; those weights are a judge-level counterpart of our personal style weights. A judge with moderate overall agreement can still rank the answers to one prompt poorly [29], and that within-request ranking is what a selection rule needs. Benchmarks published before a model's training cut-off may be in its training data, which can inflate any comparison on them [47]; guided completion of benchmark items is one way to detect this [19]. Prediction-powered inference yields valid confidence intervals from many model labels and few human labels [3]. Selective evaluation guarantees agreement with humans by abstaining when the judge is uncertain [25], and conformal risk control bounds expected loss [4]. RouteLLM trains routers on Arena preference data [38]. Implicit feedback reveals satisfaction but is noisy as a learning signal [31,33].

3 The measurement model

Setting. Fix a request, or a task type viewed as a distribution over requests. Let A be a random answer drawn from a distribution π over the answers under study, for example those of the candidate models. Let U be a random rater drawn from the population D whose judgment counts, such as the users of a product, a customer segment or a review team. Let R be an occasion that indexes repeated judgments. We assume that U, A and R are independent. A real-valued judgment is a square-integrable function $Y = f(U, A, R)$, such as a rating or the latent utility of a random-utility model. A pairwise judgment $Z(u, a, b) \in \{0, \frac{1}{2}, 1\}$ records whether u prefers a to b, with $\frac{1}{2}$ for a tie. Under the Bradley-Terry-Luce random-utility model [8,34],

$$P\big(Z(u,a,b) = 1 \mid \text{no tie}\big) \;=\; \sigma\big(U_u(a) - U_u(b)\big), \qquad \sigma(t) = 1/(1+e^{-t}). \quad (1)$$

Table 1 collects the notation.

Table 1. Notation.

Symbol	Meaning	Symbol	Meaning
a, b; A	answers; random answer	$c(a) \in \mathbb{R}^K$	checkable properties
u; U; D	rater; random rater; population	w	weights on c
r; R	occasion (repetition)	$\varphi(a) \in \mathbb{R}^F$	style and identity features
Y	real-valued judgment	$\bar{\theta}$; θ_u	population and rater style weights
$Z(u, a, b)$	pairwise judgment	$\delta_u = \theta_u - \bar{\theta}$	a rater's deviation in style taste
$U_u(a)$	rater u's utility for a	η_a; ζ_{ua}	shared and personal residuals
μ	grand mean	S	subjectivity index
α_a	shared quality	Q_m	quality of model m
β_u	leniency	κ	non-inferiority margin
γ_{ua}	personal taste	X; ω	edge flow; edge weights
ε	noise	s	scores of the gradient part

Definition 1 (Components). *With* $\mu = \mathbb{E}[Y]$,

$$\begin{aligned}
\alpha_a &= \mathbb{E}[Y \mid A = a] - \mu && \textit{(shared quality)},\\
\beta_u &= \mathbb{E}[Y \mid U = u] - \mu && \textit{(leniency)},\\
\gamma_{ua} &= \mathbb{E}[Y \mid U = u, A = a] - \mathbb{E}[Y \mid A = a] - \mathbb{E}[Y \mid U = u] + \mu && \textit{(personal taste)},\\
\varepsilon &= Y - \mathbb{E}[Y \mid U, A] && \textit{(noise)}.
\end{aligned}$$

Features. Collect the checkable properties of an answer in $c(a) \in \mathbb{R}^K$: is each claim backed by the source, is each constraint met, do the tests pass. Style and identity go in $\varphi(a) \in \mathbb{R}^F$ (length, structure, tone markers and the model that wrote it). Section 4 shows that

$$\alpha_a = w^\top c(a) + \bar{\theta}^\top \varphi(a) + \eta_a, \qquad \gamma_{ua} = (\theta_u - \bar{\theta})^\top \varphi(a) + \zeta_{ua}, \tag{2}$$

with residuals η and ζ orthogonal to the features. Substituting into Theorem 1 gives the model stated in the abstract.

Definition 2 (Subjectivity index). $S = \mathrm{Var}(\gamma) \,/\, \big(\mathrm{Var}(\alpha) + \mathrm{Var}(\gamma)\big) \in [0, 1]$*: the share of systematic, answer-dependent variation that depends on who judges.*

4 Standard results, restated

This section restates known results in the vocabulary of response quality; each carries its source. Theorem 1 is the two-way functional ANOVA [21,18], the population counterpart of the crossed random-effects model of generalizability theory [13,9]. Theorem 2 is an orthogonal projection. Theorem 3 is an immediate consequence of the Hodge decomposition of Jiang et al. [24]. Proposition 2 is Condorcet's paradox in the language of edge flows. Corollary 1 is close to a tautology, as discussed after it. Propositions 3 and 4 are elementary consequences of these results for the design of evaluations.

Theorem 1 (Orthogonal decomposition [21,18]). *$Y = \mu+\alpha_A+\beta_U+\gamma_{UA}+\varepsilon$ holds identically, the four random components are pairwise orthogonal in L^2, and*

$$\mathrm{Var}(Y) = \mathrm{Var}(\alpha) + \mathrm{Var}(\beta) + \mathrm{Var}(\gamma) + \mathrm{Var}(\varepsilon).$$

Proof. The identity is immediate: substitute Definition 1 and the terms telescope to Y. For orthogonality, put $m(u, a) = \mathbb{E}[Y \mid U = u, A = a]$. Independence of U and A does the work in this step. It gives $\mathbb{E}[m(U, A) \mid A] = \mathbb{E}[Y \mid A]$ and $\mathbb{E}[\mathbb{E}[Y \mid U] \mid A] = \mathbb{E}[\mathbb{E}[Y \mid U]] = \mu$. Hence $\mathbb{E}[\gamma_{UA} \mid A] = \mathbb{E}[Y \mid A] - \mathbb{E}[Y \mid A] - \mu + \mu = 0$, and by symmetry $\mathbb{E}[\gamma_{UA} \mid U] = 0$. So for square-integrable g, $\mathbb{E}[\gamma\, g(A)] = \mathbb{E}[g(A)\,\mathbb{E}[\gamma \mid A]] = 0$; likewise for $g(U)$. The pair α_A, β_U is easier: two independent centred variables, hence $\mathbb{E}[\alpha\beta] = 0$. Noise is easiest of all. The condition $\mathbb{E}[\varepsilon \mid U, A] = 0$ kills its inner product with any function of (U, A), including α, β and γ. Every cross term is zero and the variances add up.

Remark 1. Theorem 1 is a population statement, with raters and answers as independent draws. Holes in a crossed design are not a problem, provided what goes missing does not depend on the judgments; the estimators of Section 6 (regularised maximum likelihood with cross-validated priors) then still target these components. PRISM is another matter, because there participants wrote their own prompts. Rater and answer become dependent, the orthogonality of Theorem 1 fails, and the estimators deliver a regression decomposition instead, one that can differ from Definition 1 (Section 6.3).

Proposition 1 (Leniency cancels in comparisons). *For a single rater, $U_u(a) - U_u(b) = (\alpha_a - \alpha_b) + (\gamma_{ua} - \gamma_{ub}) + (\varepsilon_a - \varepsilon_b)$, and β does not appear in it. Under* (1)*, pairwise judgments therefore have the same distribution whatever β is, so pairwise data cannot identify β.*

Proof. The term β_u appears in both $U_u(a)$ and $U_u(b)$ and drops out of their difference.

Theorem 2 (Feature projections). *Assume the features c and φ are centred and have invertible second-moment matrices. Let $(w, \bar\theta)$ be the coefficients of the L^2 projection of α_A onto* $\mathrm{span}\{c_1(A), \ldots, c_K(A), \varphi_1(A), \ldots, \varphi_F(A)\}$*, and let $\eta = \alpha - w^\top c - \bar\theta^\top \varphi$; then $\eta \perp c, \varphi$. For each rater u, let δ_u be the coefficients of the projection of γ_{uA}, as a function of A, onto* $\mathrm{span}\{\varphi(A)\}$*, and let $\zeta_{uA} = \gamma_{uA} - \delta_u^\top \varphi(A)$; then $\zeta_{u\cdot} \perp \varphi$ for every u and $\mathbb{E}_U[\delta_U] = 0$. Writing $\theta_u = \bar\theta + \delta_u$ gives* (2)*, and $\bar\theta$ is the population average of the individual style weights.*

Proof. An orthogonal projection in a Hilbert space leaves a residual orthogonal to the target subspace. For the final claim, with $\Sigma_\varphi = \mathbb{E}[\varphi\varphi^\top]$,

$$\delta_u = \Sigma_\varphi^{-1}\mathbb{E}_A[\varphi(A)\gamma_{uA}], \qquad \mathbb{E}_U[\delta_U] = \Sigma_\varphi^{-1}\mathbb{E}_A\big[\varphi(A)\,\mathbb{E}[\gamma_{UA} \mid A]\big] = 0$$

by the proof of Theorem 1.

Remark 2. A richer set of features lets us *explain* more of α and γ, but it does not change what the model *captures*: the residuals η and ζ remain part of the shared and personal components. A new feature only shifts variation from a residual into an explained part.

Pairwise preferences and cycles. Let $G = (V, E)$ be a comparison graph on the answers with edge weights $\omega_{ab} > 0$. A skew-symmetric edge flow $X_{ab} = -X_{ba}$ has a unique orthogonal decomposition [24, Theorem 2]

$$X = \operatorname{grad} s \;\oplus\; \operatorname{curl}^* \Phi \;\oplus\; h, \qquad (\operatorname{grad} s)_{ab} = s_a - s_b, \tag{3}$$

into a gradient part, a curl part (locally cyclic) and a harmonic part (globally cyclic). The gradient part is the ω-weighted least-squares fit of one score per answer.

Theorem 3 (What utility models can represent; after [24]). *For a rater obeying* (1)*, the logit flow* $X_{ab} = \operatorname{logit} P(Z(u, a, b) = 1 \mid \text{no tie}) = U_u(a) - U_u(b)$ *is a pure gradient. Hence any curl or harmonic component in a single rater's logit flow is structure that no model of the form* (1)*, with any utility function, can represent.*

Proof. By (1), $X = \operatorname{grad} U_u$, and the decomposition (3) is unique.

Proposition 2 (Cycles from taste alone; Condorcet's paradox). *Every rater can be transitive while the pooled preferences are entirely cyclic. Take three equally weighted raters with the deterministic rankings* $a \succ b \succ c$, $b \succ c \succ a$ *and* $c \succ a \succ b$*. Then* $P(a \succ b) = P(b \succ c) = P(c \succ a) = \frac{2}{3}$*, the pooled logit flow is* $\log 2$ *on each edge of the oriented triangle, and the flow is a pure curl. The personal component represents these preferences exactly, since each rater has a utility of their own.*

Proof. Direct computation gives the probabilities. A flow that takes the same value along each edge of an oriented 3-cycle has zero divergence at every vertex, so it is orthogonal to im(grad).

Corollary 1 (Exhaustiveness relative to the measured factors). *For independent factors* (U, A, R)*, the square-integrable functions of* (U, A, R) *decompose orthogonally into eight subspaces, one for each subset of* $\{U, A, R\}$ *[21,18]. Group them into the constant;* $\{A\}$*, shared quality;* $\{U\}$*, leniency;* $\{U, A\}$*, personal taste; and every subset that contains* R*, variation across occasions, which is noise when occasions are exchangeable. Then every measurable variation of judgments over these factors splits uniquely into orthogonal components of shared quality (correctness, shared style and residual), personal taste (taste in style and residual), leniency and occasion variation; for pairwise judgments, within-rater cycles are added (Theorem 3). Consequently, no additional term built from rater, answer and occasion can exist: a proposed term is either a refinement inside one of these subspaces, such as a new feature that splits* η*, or a dependence on a factor that the model does not include.*

Proof. For independent factors, $L^2(U, A, R) = \bigoplus_{T \subseteq \{U,A,R\}} \mathcal{H}_T$, where $\mathcal{H}_T$ consists of the functions of $(X_i)_{i \in T}$ whose conditional mean given any proper subset of T is zero [18]. Theorem 1 is the stated grouping, Theorem 2 refines $\mathcal{H}_A$ and $\mathcal{H}_{UA}$, and Theorem 3 adds the only non-gradient structure of pairwise data.

Remark 3. Corollary 1 is nearly a tautology: a function of (U, A, R) is a function of (U, A, R). It makes no empirical claim. Its use is practical: a proposed new "dimension" of quality is either a feature inside one of the subspaces, which can only move variation from a residual into an explained part, or a new factor, which must be measured. Mixed logit offers a stronger expressiveness result for pairwise choice: it approximates any random-utility model [35].

Proposition 3 (What each design identifies). *Assume Gaussian random effects for the statements about variances.*

(i) From pairwise data, β is not identifiable.

(ii) With one judgment per rater-answer cell, $\mathrm{Var}(\gamma)$ *and* $\mathrm{Var}(\varepsilon)$ *are not separately identifiable, because the likelihood depends on them only through their sum. With $r \geq 2$ judgments per cell,* $\mathrm{Var}(\varepsilon)$ *is identified by the within-cell variance.*

(iii) If each answer is judged by a single rater, α_a and γ_{ua} are confounded for that answer; only the feature-explained parts $\bar{\theta}^\top \varphi$ and $\delta_u^\top \varphi$ are identifiable, by pooling across answers that share features, and the residuals η, ζ and ε remain confounded.

(iv) Within-rater cycles are identifiable only from one rater's judgments around cycles of the comparison graph.

Proof. (i) is Proposition 1. (ii) With one observation per cell, $Y_{ua} - \mu - \alpha_a - \beta_u = \gamma_{ua} + \varepsilon_{ua}$ is a sum of independent zero-mean Gaussian variables whose distribution depends only on $\sigma_\gamma^2 + \sigma_\varepsilon^2$; with replicates, the within-cell deviations $Y_{uar} - \bar{Y}_{ua\cdot}$ depend only on σ_ε^2. (iii) With a single rater per answer, $\alpha_a + \gamma_{ua}$ enters the likelihood only as a sum, while linear functionals of the features are identified by regression across answers. (iv) follows from (3), since the curl lives on cycles of the graph.

Proposition 4 (The subjectivity index). *(i) S is invariant to leniency and to noise. (ii) $S = 0$ if and only if $\gamma = 0$ almost surely, that is, all raters rank all answers alike up to leniency. (iii) $S = 1$ if and only if α is constant, that is, there is no shared quality. (iv) For independently drawn answers a, b and a common rater, the same index computed on utility differences equals S.*

Proof. (i) to (iii) follow from Definition 2 and Theorem 1. For (iv), $\mathrm{Var}(\alpha_a - \alpha_b) = 2\,\mathrm{Var}\,\alpha$ and $\mathrm{Var}(\gamma_{ua} - \gamma_{ub}) = 2\,\mathrm{Var}\,\gamma$, because $\mathrm{Cov}(\gamma_{ua}, \gamma_{ub}) = \mathbb{E}_U[(\mathbb{E}[\gamma_{UA} \mid U])^2] = 0$ by the proof of Theorem 1; the factors of two cancel.

4.1 Scope: what lies outside the model

1. *Other factors.* Context (which answers are shown together, and in which order, as in violations of independence from irrelevant alternatives), time, a rater's state and the framing of the request are further factors. Corollary 1 extends to k factors with 2^k orthogonal subspaces, so completeness is always relative to the factors measured.

2. *Aggregation.* Summarising quality for a population by α, the average, is the utilitarian choice. By Arrow's theorem [6], no aggregation of rankings over three or more options satisfies unanimity, independence of irrelevant alternatives and non-dictatorship together. The median, or the utility of the worst-off group, are equally valid summaries of the same γ. This is a decision about how to summarise γ, not a missing component.
3. *Value that no observable reflects.* Consequences that never appear in a judgment or in behaviour cannot be measured by any model. Behaviour after an answer is received (used as is, edited, regenerated) enters as a further observable linked to utility, for example $P(\text{used as is}) = \sigma(\lambda_0 + \lambda_1 U_u(a))$.

5 An accuracy-first decision rule

Let a model m answer requests x of a task type, with quality

$$Q_m = \mathbb{E}_x \, \mathbb{E}_{y \sim m(x)} \, \mathbb{E}_U[U_U(y)]$$

estimated with uncertainty. The rule is

$$\begin{aligned} m^\star &= \arg\max_m \mathrm{LB}_{0.95}(Q_m), \\ m' \text{ may replace } m^\star &\iff \mathrm{UB}_{0.95}(Q_{m^\star} - Q_{m'}) < \kappa, \end{aligned} \tag{4}$$

where $\mathrm{LB}_{0.95}$ and $\mathrm{UB}_{0.95}$ are one-sided 95% confidence bounds. Ranking by a lower bound means that a model with more evidence prevails over one that merely appears as good on less evidence. Cost enters only through the admission test, a non-inferiority test with margin κ; with $\kappa = 0$, accuracy is never traded for cost. In terms of the win rate against the incumbent, $v = P(m' \succ m^\star) + \frac{1}{2}P(\text{tie})$, non-inferiority requires the lower confidence bound of v to be at least $\frac{1}{2} - \kappa$. If a share π_d of the n comparisons is decided rather than tied, the score of one comparison (1, $\frac{1}{2}$ or 0) has variance close to $\pi_d/4$ when v is near $\frac{1}{2}$, so the required sample size is approximately $n \approx (z_{0.95} + z_{0.80})^2 \, \pi_d/(4\kappa^2)$ for 95% confidence and 80% power, that is, about 309 comparisons for $\kappa = 0.05$ and 1,930 for $\kappa = 0.02$ when half are decided ($\pi_d = 0.5$). Zero loss can never be established.

Two implications follow from Section 4. First, when the shared component carries much more than correctness, $\mathrm{Var}(\bar{\theta}^\top \varphi + \eta) \gg \mathrm{Var}(w^\top c)$, a benchmark that measures correctness alone can declare a weaker model equivalent; Section 6.1 documents such a case. Second, when S is large, Q_m conceals which raters are served well, and the rule should be applied per segment of raters, or the aggregation choice of Section 4.1 made explicit.

6 Experiments

Models and access. The experiments use Claude Opus 5.5, accessed through the Claude Code command-line client with all tools, settings and external servers disabled; DeepSeek V4.1 Flash, accessed through OpenRouter; and Gemini 3.8

Flash, accessed through Vertex AI, as a second rater family, a second fact checker and a second correctness grader. All calls were made between 24 and 26 October 2026. Model identifiers, access paths and decoding settings are listed in Table 7, and the full prompts in Appendix B.

Statistical analysis. Gains in held-out log-likelihood are paired, per judgment, between nested models. Each judgment's held-out log-likelihood is averaged over five-fold cross-validation repeated with five random fold assignments, because a single split can decide a borderline test. We report mean gains with 95% normal-approximation intervals and two-sided p-values, and we control the family-wise error rate with the Holm-Bonferroni procedure within each experiment (6 tests in the controlled study, 19 on MT-Bench). Agreement rates carry 95% bootstrap intervals from 2,000 resamples of comparisons, and the PRISM gain carries a 95% bootstrap interval from 1,000 resamples of people. The PRISM hypothesis about stated preferences was specified before the analysis was run; a second correlation is reported as exploratory.

6.1 Controlled study with simulated raters

Tasks. The study has 90 tasks, 30 of each type: documentation questions (objective), replies to support tickets (mixed) and launch posts (subjective). Sixty come from the public documentation of an AI coding agent [1]: a first set of 30, and a second set of 30 built the same way from questions and pages the first did not use. So that the results do not rest on the author's own product, the other 30 come from the documentation of uv, an open-source Python package manager [7]; uv has no question set, so Gemini 3.8 Flash, which wrote none of the answers, drafted one question or customer message per sampled passage, each from a different page.

Answers and fact checks. Every task got four answers. Claude Opus 5.5 wrote answer A and DeepSeek V4.1 Flash wrote answer B; answer C is answer A after an edit that introduced exactly one factual error, and answer D is answer B padded with filler that adds no information. Opus 5.5 checked all 360 answers against the documentation and Gemini 3.8 Flash checked them again on its own, so the correctness instrument c does not rest on the model that wrote answer A and made the error edit.

Raters. Seven raters judged every answer. One was an impartial judge; the other six were staff personas with tastes we wrote down: a terse engineer, an energetic developer advocate, an empathetic support lead, a cautious security lead, a marketer who likes punchy copy, and a manager who wants plain English. For the uv tasks the personas keep their tastes and are described as staff of the team that makes the product. Every judgment and rating was collected twice, once with the raters played by Opus 5.5 and once by Gemini 3.8 Flash.

Two designs. In the *pairwise* design, all six pairs of answers per task were judged blind, in random order, with six items per call from six different tasks, so that a call could not enforce consistency within a task; 10% of judgments were repeated in fresh calls, and DeepSeek V4.1 Flash also judged A against B. Pairwise data cannot identify leniency, and one judgment per cell cannot separate taste from noise (Proposition 3). We therefore added a crossed *rating* design: every rater rated each of the 360 answers on a 1 to 7 scale, six answers per call from six different tasks, and rated a random 10% of them again in fresh calls. Answers are nested in tasks, so we analyse the ratings with a nested random-effects ANOVA: raters are crossed with answers, answers sit within tasks, and the model adds a task effect and a rater-by-task effect to the components of Theorem 1. It gives $\mathrm{Var}(\alpha)$ and $\mathrm{Var}(\gamma)$ within a task, $\mathrm{Var}(\beta)$ and $\mathrm{Var}(\gamma) + \mathrm{Var}(\varepsilon)$; the repeats give $\mathrm{Var}(\varepsilon)$, and S within a request follows from Definition 2. This is the same quantity the pairwise design sees, since every comparison is within one task. As in the pairwise analysis, the six personas are the population and the impartial judge is an instrument, so the judge's ratings are not part of these estimates.

Fitting. For the pairwise design we fitted, per task type, nested Bradley-Terry models (Appendix A): M0, correctness only; M1, adding a shared term per answer; M2, adding personal deviations per rater and answer. Prior widths were chosen by five-fold cross-validation, and each step is credited only with its held-out log-likelihood gain. For both designs, the interval on S comes from a bootstrap over tasks that repeats the whole estimation, including the choice of prior widths, in every replicate.

Correctness and the decision. Neither fact checker found a contradicted claim in any of the 90 answers of either model, or in their padded versions; the two agree on 357 of the 360 answers, and the injected error was caught in 85 of 90 edited answers by Opus 5.5 and 84 by Gemini 3.8 Flash, never with a false flag. Personas preferred the clean answer to its one-error copy in 98 to 100% of comparisons (Table 2). The Opus judge preferred the Opus answer to the DeepSeek answer in 90 to 100% of items, the DeepSeek judge in 87 to 100%, and the Gemini judge, which wrote neither, in 90 to 97%, so the gap is not explained by self-preference alone. With identical factual records, the cheaper model's win rate against the stronger one, v with ties counted as half, was only 17 to 31% with Opus raters (lower 95% bounds 12 to 25%) and 27 to 42% with Gemini raters (21 to 35%); by documentation source it ranges from 7 to 35% and from 22 to 45%. With $\kappa = 0.05$, rule (4) requires a lower bound of at least 45%, so the cheaper model is admitted for no task type (Figure 2), although a benchmark of correctness alone would rate the two models as equal.

Shared quality and taste. In the pairwise design, shared quality beyond correctness improved held-out prediction in all three task types with raters from both families, and so did personal taste, narrowly for launch posts with Opus raters ($p_{\mathrm{Holm}} = 0.034$). Taste is visible in plain counts: in support replies the energetic advocate and the empathetic support lead preferred the padded answer in 77%

Table 2. Pairwise design: 30 tasks per type from both documentation sources, 1,080 persona judgments per type and rater family. †Significant after Holm-Bonferroni correction across the six tests of each rater family ($p_{\text{Holm}} < 0.05$). Neither fact checker found a contradicted claim in any answer A, B or D. Intervals on S come from 200 bootstrap samples of tasks; S from pairs takes values on the grid of prior widths, so its intervals are coarse, and * marks a fit at the edge of the grid.

	Documentation	Support replies	Launch posts
Raters played by Opus 5.5			
Gain from shared quality (95% CI)	$0.212 \pm 0.033^{\dagger}$	$0.103 \pm 0.033^{\dagger}$	$0.157 \pm 0.031^{\dagger}$
Gain from personal taste (95% CI)	$0.010 \pm 0.005^{\dagger}$	$0.024 \pm 0.011^{\dagger}$	$0.010 \pm 0.009^{\dagger}$
S from pairs [95% interval]	0.16 [0.08, 0.45]	0.31 [0.16, 0.44]	0.44 [0.26, 0.50]
Disagreement: between / within personas	10% / 5%	16% / 4%	13% / 6%
Cyclic triads	1/720	5/708	1/720
Clean preferred to its one-error copy	100%	98%	100%
Padded preferred to plain (personas; judge)	7%; 0%	29%; 7%	8%; 0%
Cheaper model's win rate v (lower bound)	27% (21%)	17% (12%)	31% (25%)
Raters played by Gemini 3.8 Flash			
Gain from shared quality (95% CI)	$0.142 \pm 0.032^{\dagger}$	$0.033 \pm 0.027^{\dagger}$	$0.070 \pm 0.027^{\dagger}$
Gain from personal taste (95% CI)	$0.035 \pm 0.015^{\dagger}$	$0.075 \pm 0.018^{\dagger}$	$0.085 \pm 0.021^{\dagger}$
S from pairs [95% interval]	0.36 [0.20, 0.50]*	0.50 [0.50, 0.69]	0.69 [0.50, 0.69]
Disagreement: between / within personas	18% / 6%	28% / 8%	27% / 8%
Cyclic triads	2/718	8/708	3/720
Clean preferred to its one-error copy	100%	98%	100%
Padded preferred to plain (personas; judge)	25%; 0%	34%; 50%	19%; 7%
Cheaper model's win rate v (lower bound)	27% (21%)	37% (30%)	42% (35%)
Same choice as the Opus portrayal	90%	89%	89%
Injected error caught, Opus; Gemini checker	29; 28 of 30	29; 29 of 30	27; 27 of 30

Table 3. Rating design: six personas rate all 120 answers of a type (30 tasks, four answers each) on a 1 to 7 scale, about 10% twice. Variance components from a nested random-effects ANOVA (raters crossed with answers within tasks), noise from the repeated ratings; S within a request is $\mathrm{Var}(\gamma)/(\mathrm{Var}(\alpha) + \mathrm{Var}(\gamma))$. Negative estimates are set to zero. Intervals from 2,000 bootstrap samples of tasks with the six personas fixed, so they are conditional on this population. One Gemini call failed on capacity limits until a later retry, and two items that the model returned twice were counted once. The S of 1.00 for natural answers to launch posts with Gemini raters rounds up from 0.996: the estimate of $\mathrm{Var}(\alpha)$ there is close to zero (0.01), so these raters barely agree on which of the two answers is better.

	Documentation	Support replies	Launch posts
Raters played by Opus 5.5			
$\mathrm{Var}(\alpha)$, shared quality within a task	2.22	0.89	1.17
$\mathrm{Var}(\beta)$, leniency [95% interval]	0.20 [0.14, 0.27]	0.04 [0.02, 0.06]	0.06 [0.04, 0.09]
$\mathrm{Var}(\gamma)$, personal taste [95% interval]	0.43 [0.32, 0.56]	0.31 [0.23, 0.39]	0.71 [0.56, 0.84]
$\mathrm{Var}(\varepsilon)$, noise	0.22	0.20	0.19
Task; rater × task variance	0.00; 0.00	0.00; 0.00	0.00; 0.00
S within a request [95% interval]	0.16 [0.13, 0.21]	0.26 [0.20, 0.32]	0.38 [0.31, 0.45]
S, natural answers A and B only	0.37 [0.11, 0.62]	0.53 [0.33, 0.72]	0.77 [0.68, 0.86]
S by source: 100xprompt (two sets); uv	0.19, 0.16; 0.14	0.28, 0.33; 0.16	0.34, 0.35; 0.45
Share of $\mathrm{Var}(\alpha)$ explained by correctness	44%	36%	27%
Repeated ratings identical	61%	68%	70%
Raters played by Gemini 3.8 Flash			
$\mathrm{Var}(\alpha)$, shared quality within a task	2.00	0.72	0.82
$\mathrm{Var}(\beta)$, leniency [95% interval]	1.33 [1.12, 1.58]	0.77 [0.64, 0.91]	0.35 [0.27, 0.45]
$\mathrm{Var}(\gamma)$, personal taste [95% interval]	1.31 [0.95, 1.64]	1.31 [1.08, 1.54]	2.48 [2.12, 2.83]
$\mathrm{Var}(\varepsilon)$, noise	0.55	0.45	0.38
Task; rater × task variance	0.00; 0.00	0.00; 0.00	0.00; 0.00
S within a request [95% interval]	0.40 [0.32, 0.47]	0.65 [0.55, 0.75]	0.75 [0.66, 0.84]
S, natural answers A and B only	0.63 [0.00, 0.96]	0.87 [0.73, 0.99]	1.00 [0.91, 1.00]
S by source: 100xprompt (two sets); uv	0.39, 0.35; 0.47	0.47, 0.82; 0.72	0.65, 0.64; 0.92
Share of $\mathrm{Var}(\alpha)$ explained by correctness	41%	36%	30%
Repeated ratings identical	49%	45%	42%
Correlation with the Opus portrayal's ratings	0.82	0.68	0.77

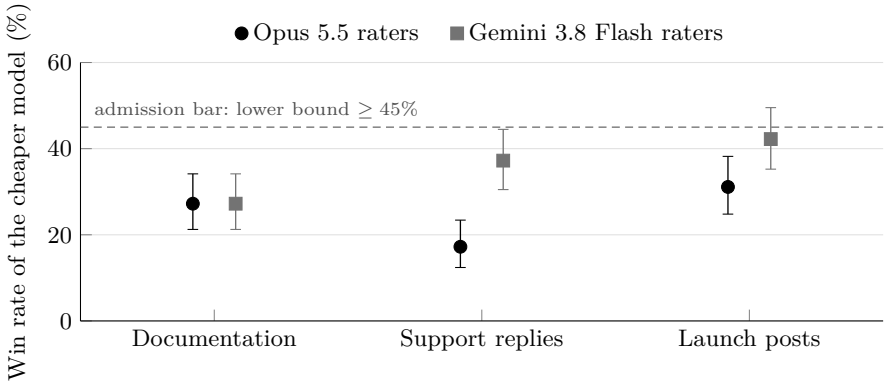


Fig. 2. The cheaper model's win rate v against the stronger one, ties counted as half, with 95% intervals, for raters played by Opus 5.5 (circles) and Gemini 3.8 Flash (squares). Rule (4) with $\kappa = 0.05$ admits the cheaper model only if the lower bound reaches 45% (dashed line). None does, although the two models have identical factual records.

and 97% of comparisons with Opus raters and in 100% with Gemini raters, while the other four personas did so in at most 3%.

Every component, from the ratings. Table 3 and Figure 3 report the rating design. Leniency, which no pairwise design can see, is clearly non-zero in every task type: $\mathrm{Var}(\beta)$ is 0.04 to 0.20 with Opus raters and 0.35 to 1.33 with Gemini raters, and no interval reaches zero. Repeated ratings give the noise, about 0.2 on the seven-point scale with Opus raters and 0.38 to 0.55 with Gemini raters, and taste exceeds it in every task type for both families. Two terms come out below zero. The moment estimate of the task variance is -0.36 to -0.05 with Opus raters and -0.39 to -0.10 with Gemini raters, and that of the rater-by-task variance -0.11 to -0.04 and -0.42 to -0.14; we set both to zero. Negative values do not mean that raters have no liking for whole tasks. They mean that the four answers of a task, which all carry the same designed contrast, are not the independent draws the model assumes. With Opus raters, S within a request is 0.16 for documentation questions, 0.26 for support replies and 0.38 for launch posts, with interval half-widths of 0.04 to 0.07, and launch posts have the highest index within each documentation source. Correctness accounts for 44% of the variance of shared quality within a task for documentation questions, 36% for support replies and 27% for launch posts. All of this is conditional. The personas are six we wrote, and half the answers were made worse on purpose (C carries an error, D is padded), which inflates $\mathrm{Var}(\alpha)$ and pulls S down. Restricted to the two models' own answers, A and B, S is 0.37, 0.53 and 0.77 with Opus raters: when every answer is acceptable, taste is most of what is left. With Gemini raters the documentation-question interval on that subset runs from 0.00 to 0.96, which tells us nothing.

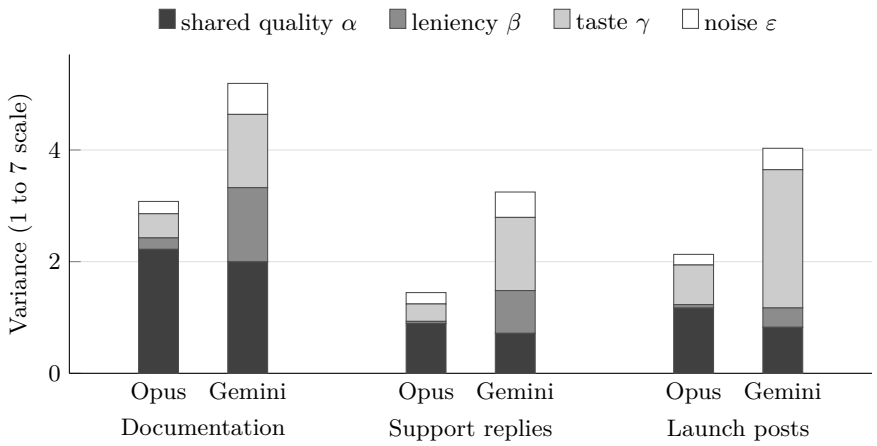


Fig. 3. Variance components of the ratings within a request, by task type and rater family, on the 1 to 7 scale: shared quality α, leniency β, personal taste γ and noise ε. The task and rater-by-task terms are not shown; their moment estimates are negative and set to zero. Gemini's personas are more lenient, noisier and express much more taste.

Scale use. S is invariant to a rater's shift but not to a rater's stretch: a persona who uses the scale more extremely while ranking answers like everyone else adds to $\mathrm{Var}(\gamma)$. Standardising each rater's ratings changes S little (0.12, 0.24 and 0.36 with Opus raters, 0.39, 0.64 and 0.73 with Gemini raters), and neither does dividing each rater's ratings by that rater's slope on the answer means (0.12, 0.25 and 0.38; 0.46, 0.66 and 0.82), although the slope correction is unstable where a persona barely tracks shared quality, as two Gemini personas do for documentation questions. Differences in how the raters use the scale do not account for the taste we measure.

What pairs can say about S. With 30 tasks per type the pairwise estimates agree with S within a request from the ratings for documentation questions and launch posts in both families (0.16 and 0.44 against 0.16 and 0.38 with Opus raters, 0.36 and 0.69 against 0.40 and 0.75 with Gemini raters); support replies match less well (0.31 against 0.26, and 0.50 against 0.65). Agreement is expected because, over all six pairs of a task's four answers, the mean squared difference of any answer-level term is exactly twice its sample variance within the task (divisor $n-1$). This identity holds for any set of answers, which matters here because C and D are edited copies of A and B, not the independent draws Proposition 4(iv) assumes. The pairwise intervals, however, are up to five times as wide, and the estimate is fragile. The cross-validated fit fixes the ratio τ/σ_q only together with the overall scale, which near-unanimous choices (a clean answer beats its one-error copy almost every time) push upward. At a fixed total width w the best S moves with w, for documentation questions with Opus raters from about 0.1 at

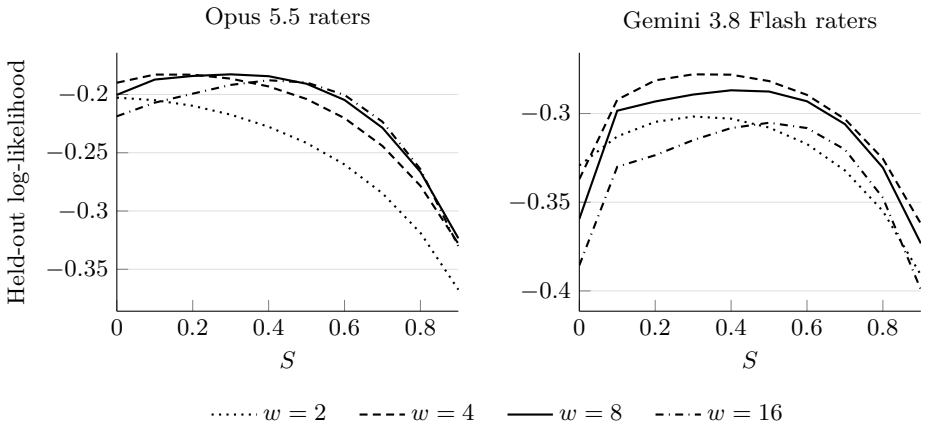


Fig. 4. Held-out log-likelihood per judgment of the pairwise model M2 for documentation questions, against S at a fixed total prior width $w = \sqrt{\sigma_q^2 + \tau^2}$. The best S moves with w, and the curves are nearly flat around their maxima, so pairwise judgments pin S down only weakly.

$w = 4$ to 0.4 at $w = 16$, and at $w = 8$ any S between 0.2 and 0.4 changes the held-out log-likelihood by less than 0.002 per judgment (Figure 4). Every pairwise S is fixed by the ratio $r = \tau/\sigma_q$ of two grid values, through $S = r^2/(1 + r^2)$: with Gemini raters the fits land on $r = \frac{3}{4}$, 1 and $\frac{3}{2}$, which give exactly 0.36, 0.50 and 0.69. The grid therefore matters. On a grid capped at $\tau = 3$, the fits to the first 30 tasks reach the boundary in four of six cases. With Opus raters S is 0.14, 0.31 and 0.31 on the capped grid and 0.45, 0.50 and 0.39 on the wider one (the first two again at its boundary), so the cap biases S downward; with Gemini raters the capped grid gives 0.20, 0.50 and 0.69, again the ratios $\frac{1}{2}$, 1 and $\frac{3}{2}$. Crossed ratings with repeats estimate S without a grid, and they are the design we recommend for it.

A second rater family. One model portraying every rater could produce structure that belongs to the model rather than to the personas, so every judgment and rating was repeated with Gemini 3.8 Flash, which wrote none of the answers (Figure 5). On identical comparisons, the Opus and Gemini versions of a persona picked the same side 89 to 90% of the time; their ratings of the same answers correlate at 0.68 to 0.82. Most of the qualitative picture survives the switch. Correct answers still win, both shared quality and taste show up in every task type, the energetic and empathetic personas still reward padding, the cheaper model is still not admitted, and documentation questions still have the lowest S. The rest of the order is shaky. Under Gemini raters, support replies (0.65 [0.55, 0.75]) and launch posts (0.75 [0.66, 0.84]) are close, and on the second set of 100xprompt tasks support replies even come out on top. The magnitudes do not replicate either. Gemini's personas disagree with each other more often (18

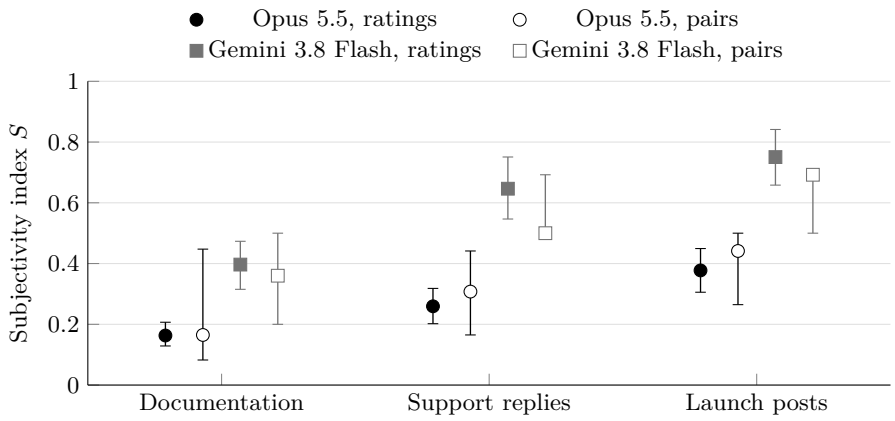


Fig. 5. The subjectivity index within a request, by task type, from crossed ratings (filled) and from pairs (open), with raters played by Opus 5.5 (circles) and Gemini 3.8 Flash (squares), and 95% bootstrap intervals over tasks. Documentation questions have the lowest index under both families; its size depends on the family, and the pairwise intervals are much wider.

to 28% of comparisons, against 10 to 16%), are more lenient and noisier, and express much more taste: S within a request is 0.40, 0.65 and 0.75 against 0.16, 0.26 and 0.38, two to two and a half times as large. The rating design has no grid, and the gap survives standardising each rater's scale, so it belongs to the portrayals, not to the estimator or to scale use. One instrument result stands out: the impartial judge played by Gemini preferred the padded answer in half of the support-reply comparisons although it was told not to reward length, which bears directly on using a judge from that family as an instrument. Estimates from simulated raters describe the portrayal as well as the personas, which is why we treat the controlled study as evidence about the estimators and not about people, and why the size of S for a real population has to be measured on that population.

6.2 MT-Bench: expert judgments

Data and fitting. MT-Bench's human judgments [53] comprise 3,355 blind pairwise votes cast by 65 experts on the answers of six 2023 models to 80 two-turn questions in eight categories; 961 comparisons were judged by two or more experts. Opus 5.5 graded the 330 answers in math, reasoning and coding that have a reference answer against MT-Bench's references, marking 105 as correct; a second grader checks these grades below. For each category we fitted nested models: L1, model strength (a leaderboard); L2, adding correctness; L3, adding shared quality (style features and a term per answer); and L4, adding each expert's personal weights on style and model features.

Table 4. MT-Bench. Held-out log-likelihood gain per vote of each added term, with 95% intervals. †Significant after Holm-Bonferroni correction across the 19 tests. Figure 6 shows how often two experts disagree in each category.

		Gain from adding		
Category	Votes	correctness	shared quality	personal taste
Writing	378	n/a	0.037 ± 0.029	0.009 ± 0.010
Roleplay	464	n/a	$0.057 \pm 0.030^{\dagger}$	0.009 ± 0.010
Humanities	452	n/a	0.017 ± 0.022	0.006 ± 0.008
STEM	416	n/a	0.003 ± 0.011	0.001 ± 0.007
Extraction	402	n/a	$0.065 \pm 0.033^{\dagger}$	0.013 ± 0.012
Reasoning	434	$0.071 \pm 0.029^{\dagger}$	$0.034 \pm 0.014^{\dagger}$	0.000 ± 0.001
Math	412	$0.119 \pm 0.031^{\dagger}$	0.004 ± 0.005	0.008 ± 0.007
Coding	397	$0.031 \pm 0.021^{\dagger}$	0.014 ± 0.011	0.002 ± 0.003

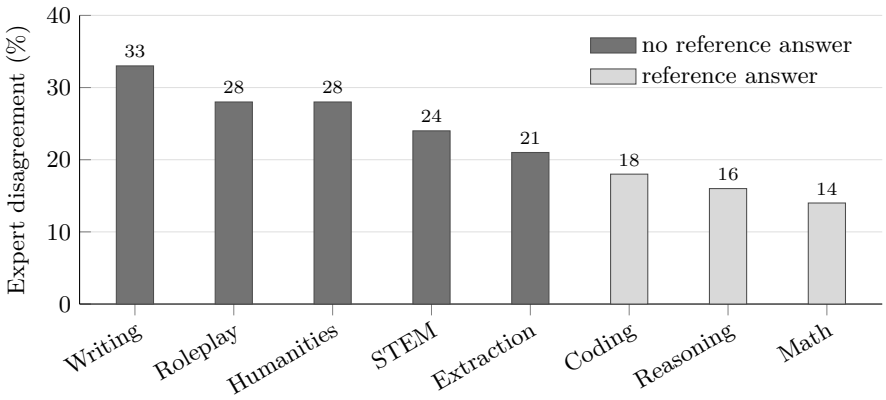


Fig. 6. How often two experts judging the same MT-Bench comparison disagree, by category. In the categories with reference answers (light) they disagree in 14 to 18% of comparisons, against 21 to 33% in those without (dark).

Results. Correctness is significant after correction in all three categories with reference answers, and dominates in reasoning and math. Among votes in which exactly one answer was correct, experts preferred it in 99% of 179 in math, 93% of 108 in reasoning and 89% of 127 in coding. These grades come from a model, so we repeated them with a grader from another family, Gemini 3.8 Flash, given the same prompt. The two graders agree on 97.6% of the 330 answers (Cohen's $\kappa = 0.94$), and on the answers both grade alike, experts prefer the correct answer in 98.6% (math), 94.2% (reasoning) and 91.2% (coding) of votes. This criterion check against expert behaviour is indirect; no grade was audited by a human. Part of the correctness signal is already carried by model strength, since the stronger models write the correct answers. Shared quality beyond model strength and correctness is significant in three of the eight categories

Table 5. Agreement with an expert's vote on the 300 MT-Bench comparisons Opus 5.5 judged. "All votes" scores A, B and tie; the leaderboard never predicts a tie, which lowers its first column. [a]Bradley-Terry scores fitted to all other comparisons in the category, excluding the comparison's own votes.

	Agreement	
Predictor of an expert's vote	all votes	both chose a side
Another expert	66.5%	83.4%
Leaderboard: stronger model of the pair, per category[a]	64.6%	83.6%
Opus 5.5 given only the question and the two model names	63.8%	81.8%
GPT-4 judge (original release)	68.0%	84.0%
Opus 5.5 judge	72.6%	86.2%
Comparisons with three or more votes (124)		
Another expert	65.7%	83.0%
Opus 5.5 judge	70.6%	85.5%
Majority of the other experts	76.6%	88.9%

(Table 4); in writing it is positive but not significant after correction ($p = 0.012$, $p_{\text{Holm}} = 0.16$), and its significance in an earlier single-split analysis did not survive repeated cross-validation. Experts disagree with each other in 14 to 18% of comparisons in math, reasoning and coding, against 28 to 33% in writing, roleplay and humanities (Figure 6). In the open-ended categories longer answers win more often, with standardised shared length weights of 0.37 to 0.65, whereas length has no effect in math and coding. Personal taste is not significant in any category. This is expected: each expert cast a median of six votes per category, too few to estimate individual weights, and without repeated judgments taste and noise are confounded (Proposition 3).

Judges compared with experts. On 300 comparisons judged by two or more experts, stratified across categories, an expert's vote agrees with another expert's in 66.5% of pairs (83.4% when both choose a side), with GPT-4's verdict from the original release in 68.0% (84.0%), and with the verdict of Opus 5.5, judging blind in random order, in 72.6% (86.2%). Opus 5.5 exceeds expert-expert agreement by 6.1 points over all votes (95% CI 1.8 to 10.1), but on votes where both choose a side the difference is 2.8 points (−0.6 to 6.3) and not significant. Table 5 sets these figures against three checks.

First, a predictor that tracks the consensus should agree with an expert more often than another expert does, because a single expert is a noisy draw around it. On comparisons with three or more votes, the majority of the other experts agrees with the held-out expert in 76.6% (88.9%), above Opus 5.5 (70.6%; 85.5%). Second, much of the signal is carried by the identity of the models alone: choosing the stronger model of the pair by a leaderboard agrees with experts in 64.6% (83.6%). Opus 5.5 exceeds it by 4.4 to 11.9 points over all votes, in

part because the leaderboard never predicts a tie, and by −0.4 to 5.8 points on votes where both choose a side. Third, contamination. Following the guided completion method of Golchin and Surdeanu [19], we gave Opus 5.5 the first half of each of the 58 MT-Bench questions with 20 or more words; it completed all 58 with the true remainder (median longest common run of 20 words for a median remainder of 20 words; the first five words exact in 54). MT-Bench is therefore in its training data, plausibly with the published model answers. Asked which answer the experts preferred, given only the question and the two model names, Opus 5.5 agreed with experts in 63.8% (81.8%), within −3.1 to +1.5 points of the leaderboard: we find no recall of individual votes beyond what the model names reveal, but memorised questions and answers could still help the judge. In sum, Opus 5.5 reproduces expert consensus somewhat better than a leaderboard and below the majority of other experts, on data it has seen. This supports its use as an instrument for shared quality, calibrated on human judgments [3]; it does not show that a model judges better than experts.

6.3 PRISM: personal taste in human ratings

Data. PRISM [28] links 1,500 participants to about six conversations each with 21 models; every response is scored from 1 to 100 by the participant who wrote the prompt. In each turn the participant sees two to four responses to the same prompt, and 1,396 participants contributed such turns (68,350 responses). Centring the scores within a turn removes both the difficulty of the prompt and the participant's leniency β_u (Proposition 1).

Fitting and control. We fitted $\bar{\theta}$ on style features and model identity by least squares across all participants, and each participant's deviation δ_u by ridge regression on the residual. Each participant's conversations were split into thirds, and each third was predicted from the other two. As a control, each participant was assigned the δ of another participant drawn at random.

Results. On conversations not used for fitting, shared quality explains 19.1% of within-turn variance and orders 62.8% of 57,907 pairs of responses correctly. Adding the participant's own taste raises these to 19.8% and 63.2%, a gain of 0.7% of variance (95% CI 0.4 to 0.9%); another participant's taste lowers the explained variance by 0.3% (95% CI 0.2 to 0.4%). Personal taste is therefore real, stable across a participant's conversations and specific to the participant, although small when expressed through coarse features: about 2% of the predicted differences in each conversation type. The shared weights favour longer responses (7.1 points per standard deviation of log length) and penalise refusals and disclaimers (−2.8 points). The pre-specified hypothesis that participants who state that safety matters to them penalise refusals less was not supported ($r = 0.044$, 95% CI −0.010 to 0.097, $n = 1{,}339$). An exploratory correlation with stated creativity ($r = -0.056$, 95% CI −0.109 to −0.003) does not survive correction for the two correlations tested.

Departure from the crossed design. PRISM participants write their own prompts and score only the responses to them, so answers are nested within participants and rater and answer are dependent. The orthogonality of Theorem 1 therefore does not hold, and what we estimate is a regression decomposition within turns (Remark 1): a participant's δ_u is their taste as expressed on prompts of their own choosing. Part of it could reflect the kind of conversation a person chose rather than how they judge. Fitting separate shared weights for each of PRISM's three conversation types (unguided, values-guided, controversy-guided) leaves the gain almost unchanged (0.6% of variance, 95% CI 0.4 to 0.9%; another participant's taste -0.3%), but finer differences between people's prompts are not controlled, and the PRISM estimate should be read as an upper bound on taste in the crossed sense.

About 80% of within-turn variation is not explained by these features. Because each PRISM response is scored once by one participant, the residuals η, ζ and ε are confounded in these data (Proposition 3(ii) to (iii)).

6.4 Preference cycles

Theorem 3 concerns cycles within one rater. MT-Bench cannot test it: no expert judged all three pairs of any three models on the same question and turn (0 within-expert triads; each expert cast at most four votes per question and turn). What the data allow is a test of cycles in *pooled* preferences, which by Proposition 2 may arise from differences in taste alone and are represented by the model.

For each question and turn we formed the comparison graph on the six models, with flows $X_{ij} = (\text{wins}_i - \text{wins}_j)/n_{ij}$ and weights n_{ij}, and measured the cyclic share of the flow's energy, $1 - \|\operatorname{grad} \hat{s}\|_\omega^2 / \|X\|_\omega^2$. The graphs are sparse (1.85 votes per edge, 17 experts per graph). Finite samples produce cycles even from transitive preferences, so each graph was compared with a parametric bootstrap from a Bradley-Terry fit with the graph's vote counts and tie rate (200 draws). The observed cyclic share is 20.1%, against 30.7% under the null; no graph lies in the upper 5% tail of its null. The null is, however, miscalibrated: 21% of graphs lie in its lower 5% tail, where 5% are expected, and this persists when the ridge penalty on the fitted scores is reduced a hundredfold (null 28.1%, 20% of graphs in the lower tail). The likely causes are that the simulation spreads ties uniformly over edges, whereas real ties concentrate on close pairs, and that the votes of one expert within a graph are not independent. A tie model such as Davidson's [14] and a null that resamples experts would address both. We therefore draw no conclusion about intransitivity from MT-Bench: the test finds no excess pooled cyclicity, but its null is not trustworthy, and it says nothing about cycles within a rater. In the controlled study, where each simulated rater judged complete triads, 7 of 2,148 triads with Opus raters and 13 of 2,146 with Gemini raters were cyclic (Table 2).

Table 6. Evidence by component and source. Every component was estimated with simulated raters; only correctness, shared quality and personal taste were tested on human judgments.

Component	Controlled study (simulated raters)	MT-Bench (65 experts)	PRISM (1,396 participants)
Correctness $w^\top c$	Injected error caught in 85/90 and 84/90 by two checkers; clean answer preferred in 98 to 100%	Correct answer preferred in 89 to 99% of votes where one is correct; significant in all three reference categories	Not measured
Shared quality	Significant in all three task types with raters from both families	Significant in 3 of 8 categories	19% of held-out within-turn variance
Personal taste	Significant in all three task types with both families; exceeds noise in the ratings of both families	Not identifiable: a median of six votes per expert and category, no repeats	Detected but small (0.6 to 0.7%); may include differences in chosen prompts
Leniency β	Estimated from ratings: Var(β) 0.04 to 0.20 (Opus) and 0.35 to 1.33 (Gemini), all above zero	Cancels in pairwise data	Removed by centring; not estimated
Noise ε	Var(ε) about 0.2 (Opus) and 0.4 to 0.55 (Gemini) from repeated ratings; repeated choices differ in 4 to 6% (Opus) and 6 to 8% (Gemini)	Not identifiable: no repeats	Not identifiable: no repeats
Within-rater cycles	7 of 2,148 (Opus) and 13 of 2,146 (Gemini) triads cyclic	Not testable: no within-expert triads	Not testable: ratings, not choices

6.5 Summary: what each source can test

Table 6 collects, for each component of the model, what each source established and what its design leaves untestable.

6.6 What a human panel would need

The open questions are about people, and Proposition 3 says which design can answer them: a fixed panel that rates a fixed set of answers, rates them twice, and judges complete triads. To size it we simulated such panels from the nested model of Section 6.1, drawing new raters and new tasks for every simulated panel,

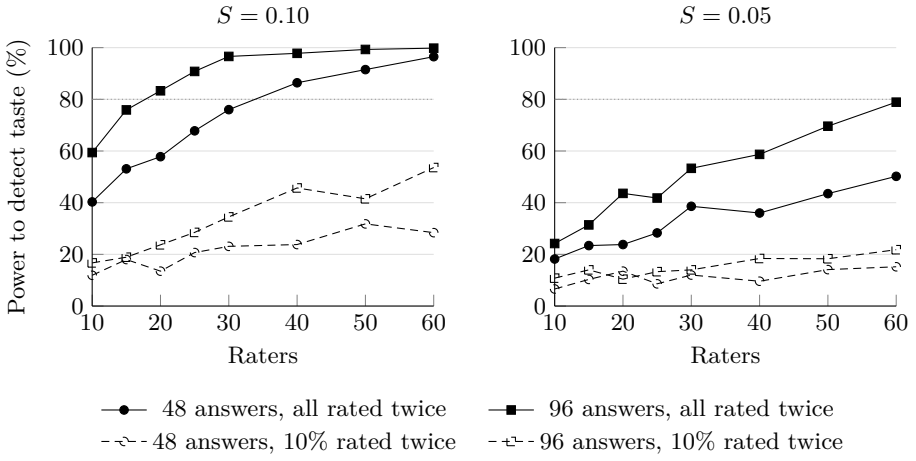


Fig. 7. Simulated power to detect personal taste in a human panel under the conservative scenarios, with $S = 0.10$ (left) and $S = 0.05$ (right), by number of raters, for 12 or 24 tasks (48 or 96 answers) with every answer rated twice (solid) or 10% of answers rated twice (dashed). Each point is 1,000 simulated panels with new raters and tasks; the dotted line marks 80%.

with the variance components of Table 3 and with two conservative scenarios in which S within a request is only 0.05 or 0.10 and the noise variance equals $\mathrm{Var}(\alpha)$, far noisier than either simulated rater family (1,000 panels per design; code in the supplement). Taste is tested with an F test of the rater-by-answer mean square against the noise variance from the repeats, and leniency with the rater mean square against the rater-by-task mean square, both at the 5% level with critical values simulated under the null (Figure 7). Leniency was detected in at least 89% of panels in every design. With the components of Table 3, even 10 raters rating 48 answers detect taste, but in the conservative scenarios the share of repeated ratings matters more than the number of raters. Homan et al. [22] find a similar trade-off when systems are compared: for a fixed total number of responses, fewer items with more responses each can make significance tests more reproducible. With 10% of ratings repeated, 25 raters rating 96 answers detect taste at $S = 0.10$ in only 27% of panels. With every answer rated twice, 40 raters detect it in 87% of panels when they rate 48 answers and in 99% when they rate 96, with $\hat{S}$ between 0.05 and 0.16 in 95% of the larger panels. We therefore propose at least 40 raters and 12 tasks with four answers each, better 24 tasks, every answer rated twice, and all six pairs of each task judged by every rater for the cycle test, collected after the training cut-off of any model used as an instrument. Such a panel is powered for $S = 0.10$, not for $S = 0.05$: at $S = 0.05$ even the larger version detects taste in only 58% of panels. Because its raters are a sample, its intervals must resample raters as well as tasks.

7 Discussion

Within its scope, the framework turns "subjective" into quantities that vary by task: in the controlled study, crossed ratings put the subjectivity index within a request at 0.16 for documentation questions and 0.38 for launch posts with Opus raters (0.40 and 0.75 with Gemini raters), and expert disagreement on MT-Bench rises from 14 to 18% in categories with reference answers to 21 to 33% in those without. Its most useful consequences are about design (Proposition 3). Measuring taste requires several raters per answer and many answers per rater; separating taste from noise requires repeated judgments; seeing leniency requires ratings rather than choices; detecting intransitivity requires that a rater judge complete triads; and estimating taste in the crossed sense requires that raters do not choose what they judge. To these the controlled study adds one more: when most choices are unanimous, pairwise judgments estimate S only weakly, so S should be measured with crossed ratings and repeats. Dsouza and Kovatchev [16] likewise find no clear advantage of preference labels over single ratings in instruction-tuning data, with disagreement depending mainly on the labelling format and the type of annotator. Neither MT-Bench nor PRISM meets these requirements, which is why Table 6 has as many untestable cells on human data as tested ones. The controlled study shows that a design meeting all of them does estimate every component, and Section 6.6 sizes it for people.

For model selection, rule (4) uses correctness checks for $w^\top c$, blind pairwise preference for shared quality, and analysis by segment when S is large, with a cheaper model admitted only on evidence of non-inferiority. The controlled study shows why correctness alone is not enough: two models with identical factual records were separated by raters from both model families. The cheaper model won 17 to 42% of comparisons, and every 95% upper bound stays below one half, though only just (0.495) for launch posts with Gemini raters.

Whether a strong model can replace human raters is not settled by our data. On MT-Bench, Opus 5.5 agreed with experts somewhat more often than experts agreed with each other, but a predictor that tracks the consensus should do so by construction, and part of the agreement is available from the identity of the models alone (Section 6.2). A model judge is best treated as an instrument for shared quality that must be calibrated on a human sample [3]; that sample is also the only source of the specific population's taste, and the only check on a judge's preference for its own model family [40], which rule (4) would otherwise convert into cost.

8 Limitations

1. *Simulated raters.* The personas are tastes we wrote, portrayed by language models. The controlled study shows what the estimators recover and that its qualitative conclusions hold across two rater families, two documentation sources and two designs. The size of S does not: it is two to two and a half times as large when Gemini rather than Opus portrays the raters. The study

does not show that people have this structure. We did not run a human panel. Section 6.6 gives the design and size one would need.

2. *Coverage of human data.* Leniency, noise and within-rater cycles were not tested on human judgments, and personal taste was detected only on PRISM, where it is small and may include differences in the prompts people chose (Table 6).
3. *Instruments.* Correctness on MT-Bench and the fact checks of the controlled study were graded by language models. Graders from two families agree on 97.6% of MT-Bench answers and on 357 of 360 controlled-study answers, and experts strongly prefer the answers graded correct, but no grade was audited by a human.
4. *Contamination.* MT-Bench predates the training of every model we used as an instrument. The probes in Section 6.2 bound, but cannot exclude, an effect of memorisation on the comparison of Opus 5.5 with experts; human judgments collected after the judge's training cut-off would remove the concern.
5. *Cycle test.* The MT-Bench test concerns pooled cycles only, and its null distribution is miscalibrated (Section 6.4).
6. *Sample and source.* The controlled study has 30 tasks per task type; intervals on S from the ratings have half-widths of 0.04 to 0.10, and those from pairs are much wider. Two thirds of its tasks come from the author's product; results are reported for each source.
7. *Scope.* Exhaustiveness holds relative to the measured factors and is close to definitional (Corollary 1).

9 Conclusion

Standard variance-components results give a precise vocabulary for the quality of an answer to a request without a single right answer: what raters share, what depends on the rater, leniency, noise and, for pairwise judgments, within-rater cycles. The vocabulary is useful mainly because it says which data can measure which part. On existing human data, correctness and shared quality are clearly present, personal taste is real but small through coarse features, and the remaining components cannot be tested with the designs available. With simulated raters every component can be estimated, and the subjectivity index is lowest for documentation questions under both rater families, but its size depends on which model portrays the raters, and only crossed ratings with repeats estimate it reliably. A panel of about 40 people rating 48 to 96 answers twice could test the open questions about humans for taste as small as $S = 0.10$; Section 6.6 gives its design.

Ethics statement

This work collected no new data from human participants. It analyses two public datasets released by their authors, PRISM, whose participants gave informed

consent [28], and MT-Bench's expert judgments [53], under their licences (see the data availability statement). The "staff personas" in Section 6.1 are simulated by language models and represent no real person, including where they are described as staff of the uv team. A subjectivity index can be misused to justify serving only the majority taste; because summarising quality by the population average is a choice (Section 4.1), we recommend reporting quality per segment of raters wherever S is non-negligible. LLM judges carry known biases [53,17], and their use as instruments should be calibrated against human judgments.

Data and code availability

MT-Bench human judgments: Hugging Face dataset `lmsys/mt_bench_human_judgments`, revision `f7d2896`, CC-BY-4.0. MT-Bench questions and reference answers: FastChat repository, file `fastchat/llm_judge/data/mt_bench/question.jsonl`, commit `b494d0c`, Apache-2.0. PRISM: Hugging Face dataset `HannahRoseKirk/prism-alignment`, revision `18ab5cf`; human-written text is licensed CC-BY-4.0 and model responses CC-BY-NC-4.0, and our use of the responses is limited to non-commercial research analysis. uv documentation: `github.com/astral-sh/uv`, directory `docs`, commit `656d02b`, MIT and Apache-2.0. All analysis code, the controlled study's tasks (with the documentation passages used), answers, fact checks from both checkers, and pairwise judgments and ratings from both rater families, and our derived MT-Bench outputs (grades from both graders, judge verdicts and probe responses) are provided as supplementary material (`supplement.zip`), with instructions for obtaining the two public datasets, which we do not redistribute.

Competing interests

The author is the CEO of 100xprompt, which develops AI coding agents and model-routing software that could benefit from the evaluation methods described here. Sixty of the 90 tasks of the controlled study were built from 100xprompt's public product documentation; the other 30 use the documentation of an unrelated open-source project, and results are reported for each source.

Use of AI assistants

The author used Claude (Anthropic) to help write the experiment code, analyse the data, and draft and edit the text; the author checked all of it and takes full responsibility for the paper. Claude Opus 5.5, DeepSeek V4.1 Flash and Gemini 3.8 Flash were also used as instruments in the experiments, as described in Section 6.

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A Fitting details and hyperparameters

Pairwise models. We fitted maximum a posteriori Bradley-Terry models by Adam (learning rate 0.05; 1,500 iterations in the controlled study, 600 on MT-Bench) with Gaussian priors. Ties enter as half wins in the log-likelihood $y \log \sigma(\Delta) + (1 - y) \log(1 - \sigma(\Delta))$. In the controlled study, the shared term per answer has prior standard deviation σ_q, chosen from $\{0.5, 1, 2, 3, 5, 8\}$ for M1 and then refined jointly with the prior standard deviation of the personal deviations, $\tau \in \{0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 9\}$, over $\{\frac{1}{2}, 1, \frac{3}{2}, 2\}$ times that value for M2, by five-fold cross-validation over judgments on a fixed split; the reported gains then average five random five-fold splits. M0 uses the mean of the two fact checkers' counts of contradicted claims. The capped grid of Section 6.1 is $\sigma_q \in \{0.5, 1, 2, 3, 5\}$, $\tau \in \{0.25, 0.5, 1, 1.5, 2, 3\}$ and multipliers $\{\frac{1}{2}, 1, \frac{3}{2}\}$. Intervals on S resample the tasks of a type with replacement (200 replicates for the pairwise design, 2,000 for the ratings) and repeat the selection of the prior widths in every replicate.

Ratings. For each task type, the six personas' first ratings of the 30 tasks $\times$ 4 answers form a complete table. With $n_u = 6$ raters, $n_t = 30$ tasks and $n_a = 4$ answers per task, the mean squares for tasks, answers within tasks, raters, rater $\times$ task and rater $\times$ answer within task (MS_T, MS_A, MS_U, MS_{UT}, MS_{UA}) give $\mathrm{Var}(\gamma) = MS_{UA} - \mathrm{Var}(\varepsilon)$, $\mathrm{Var}(\alpha) = (MS_A - MS_{UA})/n_u$, $\mathrm{Var}(\beta T) = (MS_{UT} - MS_{UA})/n_a$, $\mathrm{Var}(\beta) = (MS_U - MS_{UT})/(n_t n_a)$ and $\mathrm{Var}(T) = (MS_T - MS_A - MS_{UT} + MS_{UA})/(n_u n_a)$, with $\mathrm{Var}(\varepsilon) = \frac{1}{2}\overline{(Y_1 - Y_2)^2}$ from the repeated ratings. Negative estimates are set to zero. The bootstrap keeps the six personas fixed, since they are the population, and resamples tasks together with their

repeats; for a human panel, whose raters are a sample, it must resample raters as well. For the scale-use checks, each rater's ratings (and repeats) are either standardised by that rater's mean and standard deviation, or centred and divided by the slope of a regression of that rater's ratings on the answer means; both are recomputed in every bootstrap replicate.

MT-Bench. On MT-Bench the model comprises model strengths (prior standard deviation 2), a correctness weight (5), shared style weights (2) and a term per answer ($\sigma_\eta \in \{0.5, 1, 2\}$), and per-expert weights on the same style features and on model indicators ($\tau \in \{0.1, 0.25, 0.5, 1\}$), with all selection and all reported gains on the repeated splits. The style features are the logarithm of the length in characters, the number of markdown items (headers, list items and bold spans) and the presence of a code block, standardised within category. Intervals on gains are ± 1.96 standard errors of the per-vote differences in held-out log-likelihood.

PRISM. The features are the standardised style features above, indicators that the response ends with a question and that it contains a refusal or a disclaimer, and indicators of the model. Shared weights were fitted by ridge regression ($\lambda = 1$) and per-participant weights by ridge regression on the residuals, with $\lambda \in \{3, 10, 30, 100, 300\}$ selected by held-out variance explained ($\lambda = 100$).

Table 7. Models, access and decoding settings.

Model	Identifier and access	Settings
Claude Opus 5.5	`claude-opus-5-5`; Claude Code command-line client 2.1.309	effort "medium"; default sampling; tools, settings and external servers disabled
DeepSeek V4.1 Flash	`deepseek-v4.1-flash-1017`; OpenRouter, hosts Fireworks and InferenceNet	temperature 0.3; reasoning enabled
Gemini 3.8 Flash	`google/gemini-3.8-flash`; Vertex AI, global endpoint (OpenAI-compatible)	reasoning effort "medium"; default sampling

Seeds and software. Random seeds: task sampling and presentation order 20261025 (first 30 tasks and MT-Bench); the 60 further tasks and the text order of the second fact check of the first 30, 20261026; text order of the fact checks of the 60 tasks, 20261027; sides in pairwise judgments, 20261028 (Opus raters) and 20261029 (Gemini raters); rating batches, 20261030 and 20261031; bootstraps of S, 7 (pairs) and 11 (ratings); panel simulations, one seed per scenario and design derived from 20261101; control assignment 7 and bootstrap 11 (PRISM); cycle-test bootstrap 3. Folds were assigned deterministically by index. Software: Bun 1.3.13 and Tectonic 0.17.0. At list prices, the first 30 tasks of the controlled

study cost about $28 in model calls and their replication with Gemini raters about $6; the extension (60 tasks, the second fact checks, both rater families and the ratings of all 360 answers) cost about $93 for Opus 5.5, $15 for Gemini 3.8 Flash (part of it estimated from call counts) and $0.11 for DeepSeek V4.1 Flash; MT-Bench grading and judging cost $3.79 and the MT-Bench checks (probes and second grader) about $1.30; the PRISM analysis and the cycle test ran offline.

B Prompts

Text in angle brackets is substituted per item. Documentation passages are inserted verbatim.

Answer generation (A and B).

```
<request>

Use only the facts in these passages from 100xprompt's documentation:

<passages>

Reply with the answer text only.
```

The three request templates were: *A user of 100xprompt (a terminal AI coding agent) asks: "<question>". Answer the question.*; *You work on 100xprompt's support team. A customer wrote in: "<message>". Write the reply email to this customer.*; and *Write a short LinkedIn post (under 150 words) from the 100xprompt team announcing this capability to developers: "<feature>".* For the uv tasks, 100xprompt reads uv in this prompt and in the fact check, and the templates begin *A user of uv (a Python package and project manager) asks*, *You work on uv's support team* and *from the uv team.*

One-error edit (C).

```
Here is a text and the documentation it is based on.
DOCUMENTATION: <passages>   TEXT: <answer A>
Edit the text to introduce exactly ONE factual error that contradicts the
↪  documentation: change one specific detail (a key, command, flag,
↪  number, name or behaviour) to something plausible but wrong. Change
↪  nothing else at all: same wording, length and format everywhere else.
↪  Reply with the edited text only.
```

Padding edit (D).

```
TEXT: <answer B>
Add two or three sentences of generic filler to this text (restating what
↪  is already there, generic encouragement or pleasantries) that add NO
↪  new information. Keep every existing sentence exactly as it is. Reply
↪  with the padded text only.
```

Fact check.

```
You are a fact checker. DOCUMENTATION (the only ground truth): <passages>
TEXT 1: ...  TEXT 4: <the four answers, shuffled>
For each text, list every specific factual claim it makes about
↪  100xprompt (commands, keys, flags, behaviours, numbers, names; not
↪  opinions or pleasantries), and judge each against the documentation:
↪  "supported", "contradicted" (the documentation says otherwise) or
↪  "unsupported" (the documentation does not say). Reply with JSON only.
```

Pairwise judgment.

```
<rater description>

You will review 6 unrelated items. Each has a request, the documentation
↪  it must rely on, and two responses written for it. For each item, say
↪  which response you prefer (or "tie" only if you truly cannot choose),
↪  and whether you would approve each response for use exactly as it is.
=== ITEM k === REQUEST: ... DOCUMENTATION: ... RESPONSE 1: ... RESPONSE
↪  2: ...
Reply with JSON only: [{"item": 1, "preferred": 1 | 2 | "tie",
↪  "use_as_is": [true|false, true|false]}, ...]
```

Rater descriptions.

```
judge: You are an impartial expert reviewer. Judge which response better
↪  serves the request: correct against the documentation, helpful, clear
↪  and appropriate for its purpose. Do not let length or formatting sway
↪  you unless they genuinely help the reader.
priya: You are Priya, a staff backend engineer at 100xprompt. You value
↪  precision and brevity above all: exact commands and keys, no fluff,
↪  no marketing tone, no exclamation marks. Padding and pleasantries
↪  annoy you.
marco: You are Marco, 100xprompt's developer advocate. You love friendly,
↪  energetic writing with a concrete example, a bit of personality and a
↪  clear call to action. Dry, terse text feels cold to you.
aisha: You are Aisha, 100xprompt's support team lead. You care that the
↪  reader feels heard: acknowledge the problem, then clear numbered
↪  steps and a next step if it does not work. Curt answers feel
↪  dismissive to you.
```

```
tom: You are Tom, 100xprompt's security and compliance lead. You distrust
↪  overclaiming and hype. You want careful, formal wording, caveats
↪  where they belong and nothing promised that the documentation does
↪  not promise.
lena: You are Lena, 100xprompt's product marketing manager. You want
↪  punchy, benefit-led, scannable text with a strong hook in the first
↪  line. Long paragraphs and caveats lose readers, in your view.
kenji: You are Kenji, an engineering manager at 100xprompt whose first
↪  language is Japanese. You want plain English: short sentences, the
↪  bottom line first, no idioms or slang.
```

For the uv tasks the personas keep their tastes but not the company: *at 100xprompt* reads *on the team that builds this product*, and *100xprompt's developer advocate* (and each other role) reads *the developer advocate for this product*.

Rating (crossed design).

```
<rater description>

You will rate 6 unrelated responses. Each has a request, the
↪  documentation it must rely on, and one response written for it. Rate
↪  each response from 1 (very poor) to 7 (excellent) for how well it
↪  serves the request, by your own standards.
=== ITEM k === REQUEST: ... DOCUMENTATION: ... RESPONSE: ...
Reply with JSON only: [{"item": 1, "rating": 1 | 2 | 3 | 4 | 5 | 6 | 7},
↪  ...] with one entry per item.
```

Questions for the uv tasks (Gemini 3.8 Flash).

```
Here is a passage from the documentation of uv, a Python package and
↪  project manager.
PASSAGE: <passage>
<one of:>
Write one question a uv user might ask whose answer is a specific fact
↪  stated in this passage (a command, flag, setting, file name, default
↪  or behaviour). The question must be answerable from the passage
↪  alone.
Write one "how do I ..." question a uv user might ask that this passage
↪  answers with concrete steps or commands. The question must be
↪  answerable from the passage alone.
Write the message a uv user might send to uv's support team about a
↪  problem that this passage resolves: two to four sentences, in the
↪  user's own words, describing what they tried and what went wrong. Do
↪  not quote or mention the documentation.
Reply with JSON only: {"question": "..."}
```

MT-Bench correctness grading.

```
Grade answers to a <category> question against the reference answer.
↪  Judge only whether each answer's final result is correct (for code:
↪  whether it correctly does what was asked); ignore style and length.
CONVERSATION: <user turns>   REFERENCE ANSWER for turn <t>: <reference>
ANSWER 1 (to turn <t>): ...   <the six models' answers, shuffled>
Reply with JSON only: [{"answer": 1, "correct": true}, ...] with one
↪  entry per answer.
```

MT-Bench judging.

```
You are judging AI assistants' answers. For each item, read the
↪  conversation and decide which assistant's answer to the LAST user
↪  turn is better: more helpful, relevant, accurate, deep and well
↪  reasoned. Do not let length or order sway you. Answer "A", "B" or
↪  "tie".
=== ITEM k === <conversation>  ASSISTANT A: ...  ASSISTANT B: ...   (six
↪  items per call, random order)
Reply with JSON only: [{"item": 1, "verdict": "A" | "B" | "tie"}, ...]
```

Contamination probe: question recall.

```
These are the openings of questions from the MT-Bench benchmark (LMSYS,
↪  Zheng et al., 2023). For each, if you remember the question, continue
↪  it word for word to its end; if you do not, write the most likely
↪  continuation. Reply with JSON only: [{"item": 1, "continuation":
↪  "..."}, ...]
ITEM k (category: <category>): <first half of the words of turn 1> ...
↪  (ten items per call)
```

Contamination probe: verdict recall from model names.

```
MT-Bench (LMSYS, Zheng et al., 2023) collected expert human votes between
↪  the answers of six chat models (gpt-4, gpt-3.5-turbo, claude-v1,
↪  vicuna-13b-v1.2, alpaca-13b, llama-13b) to two-turn questions. For
↪  each item below you get the conversation up to the judged turn and
↪  the two models whose answers to its last turn were compared, but not
↪  the answers. Say which model's answer the human experts preferred,
↪  "A" or "B", or "tie". If you remember the actual votes, use them;
↪  otherwise give your best guess.
=== ITEM k (category: <category>) === <conversation>  MODEL A: <name>
↪  MODEL B: <name>   (15 items per call, random order)
Reply with JSON only: [{"item": 1, "verdict": "A" | "B" | "tie"}, ...]
```

Gemini 3.8 Flash received the MT-Bench grading prompt and the fact-check prompt above unchanged as the second grader, and as the second rater family it received the judgment and rating prompts and the rater descriptions above unchanged.